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**Java SE 27 & JDK 27**

# Class FileDescriptor

java.lang.Object
    java.io.FileDescriptor

---

```
public final class FileDescriptor
extends Object
```

Instances of the file descriptor class serve as an opaque handle to the underlying machine-specific structure representing an open file, an open socket, or another source or sink of bytes. The main practical use for a file descriptor is to create a `FileInputStream` or `FileOutputStream` to contain it.

Applications should not create their own file descriptors.

**Since:**

1.0

## *Field Summary*

**Fields**

| Modifier and Type | Field | Description |
|---|---|---|
| `static final FileDescriptor` | **`err`** | A handle to the standard error stream. |
| `static final FileDescriptor` | **`in`** | A handle to the standard input stream. |
| `static final FileDescriptor` | **`out`** | A handle to the standard output stream. |

## *Constructor Summary*

**Constructors**

| Constructor | Description |
|---|---|
| **`FileDescriptor`**`()` | Constructs an (invalid) FileDescriptor object. |

## *Method Summary*

**All Methods** | **Instance Methods**

| Modifier and Type | Method | Description |
|---|---|---|

| | | |
|---|---|---|
| void | **sync**() | Force all system buffers to synchronize with the underlying device. |
| boolean | **valid**() | Tests if this file descriptor object is valid. |

**Methods declared in class Object**

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

## *Field Details*

### in

```
public static final FileDescriptor in
```

A handle to the standard input stream. Usually, this file descriptor is not used directly, but rather via the input stream known as `System.in`.

**See Also:**

`System.in`

### out

```
public static final FileDescriptor out
```

A handle to the standard output stream. Usually, this file descriptor is not used directly, but rather via the output stream known as `System.out`.

**See Also:**

`System.out`

### err

```
public static final FileDescriptor err
```

A handle to the standard error stream. Usually, this file descriptor is not used directly, but rather via the output stream known as `System.err`.

**See Also:**

`System.err`

## *Constructor Details*

### FileDescriptor

```
public FileDescriptor()
```

Constructs an (invalid) FileDescriptor object. The fd or handle is set later.

## *Method Details*

### valid

```
public boolean valid()
```

Tests if this file descriptor object is valid.

**Returns:**

`true` if the file descriptor object represents a valid, open file, socket, or other active I/O connection; `false` otherwise.

### sync

```
public void sync()
          throws SyncFailedException
```

Force all system buffers to synchronize with the underlying device. This method returns after all modified data and attributes of this FileDescriptor have been written to the relevant device(s). In particular, if this FileDescriptor refers to a physical storage medium, such as a file in a file system, sync will not return until all in-memory modified copies of buffers associated with this FileDescriptor have been written to the physical medium. sync is meant to be used by code that requires physical storage (such as a file) to be in a known state For example, a class that provided a simple transaction facility might use sync to ensure that all changes to a file caused by a given transaction were recorded on a storage medium. sync only affects buffers downstream of this FileDescriptor. If any in-memory buffering is being done by the application (for example, by a BufferedOutputStream object), those buffers must be flushed into the FileDescriptor (for example, by invoking OutputStream.flush) before that data will be affected by sync.

**Throws:**

`SyncFailedException` - Thrown when the buffers cannot be flushed, or because the system cannot guarantee that all the buffers have been synchronized with physical media.

**Since:**

1.1

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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